

CTL7010 Ethical Research Writing in Automotive Engineering

Dr. Gokulnath Thandavarayan

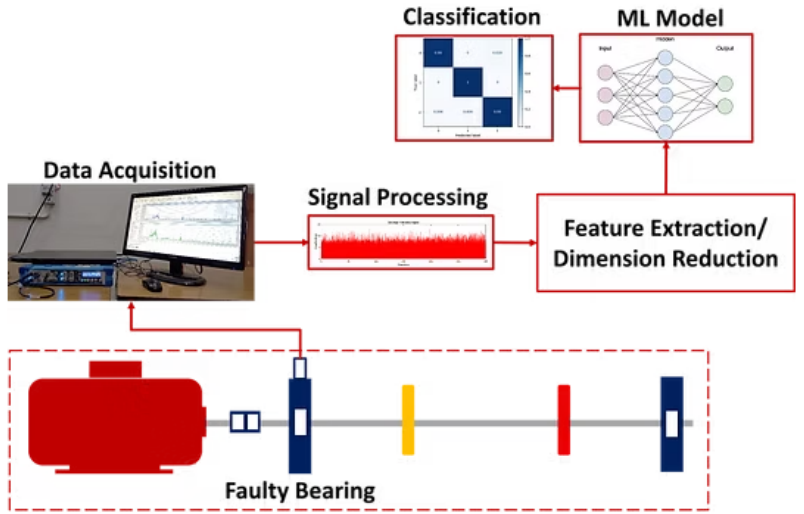
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How to Write a Research Paper

- Study Design and Validation Methods
  - Experimental test: Physical testing setup to validate models or observe real-world behavior
  - Simulation test: Virtual testing using simulation tools to analyze system dynamics
  - Theoretical algorithm test: Analytical validation using equations, proofs, or derived models
- Example: Experimental Validation Setup
  - Sensors collect raw data through a data-acquisition system.
  - Signals are processed and features extracted for classification
  - A machine-learning model predicts system behavior (e.g., faulty bearing detection)





How to Write a Paper: Simulators

- Simulation Test: Event driven or Time driven

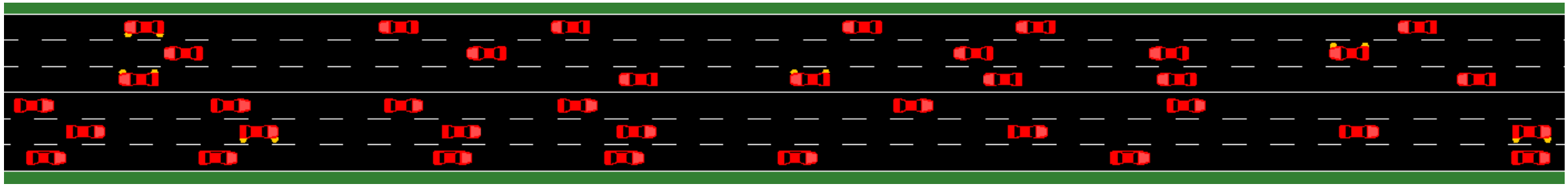
- **Event driven:** ns-3, OMNeT++
  - Maintains a **future event list (FEL)**
  - Vehicle A transmits a packet at t = 1.200 s.
  - Vehicle B receives it at t = 1.202 s (propagation + processing delay).
  - Simulation jumps from 1.200 → 1.202 s directly.
  - No computation between 1.200 s and 1.202 s.
  - Efficient because time only moves when an actual event happens.

- **Time-Driven Simulation:** Simulink, MATLAB
  - The simulation might advance every 1 ms ($\Delta t = 0.001$ s).
  - It checks system state at 1.200, 1.201, 1.202, ... even if nothing happens most of the time.
  - Inefficient if events are sparse, simpler to model continuous changes like motion, temp

How to Write a Paper: Event driven Simulators

- Simulation Workflow
  - Traffic simulator: Generates realistic vehicle movements and traces
    - Example: **SUMO** – provides mobility traces such as position, speed, and trajectory
  - Network simulator **:** Models wireless communication between vehicles
    - Example: **ns-3**, **OMNeT++** – simulate V2X message exchange and channel behavior

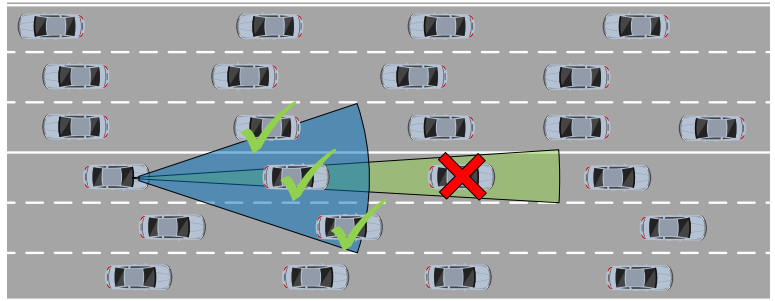
- Integration Process
  - The traffic simulator output (vehicle traces) is fed into the network simulator
  - Essential for studying evaluation of mobility patterns and network performance



How to Write a Paper: Simulation configurations

- Highway scenario: 5km
  - High traffic density ➲ 240veh/km, Speed ➲ 50Km/h, Number of lanes ➲ 8
- Sensor configuration:
  - 360º sensor: single sensor with 150m range and ±360 degrees
  - Sensors masking ➲ detect only the vehicles that are in their Line-of-Sight (LOS)
- PHY Layer: IEEE 802.11p radio access technology

Comms. parameter	Values
Carrier frequency	5.9 GHz
Channel bandwidth	10 MHz
Transmission power	23 dBm
Energy detection threshold	-85 dBm
Data rate	6 Mbps (QPSK 1⁄2)
Radio propagation	Winner+ B1 propagation model



- **Key Takeaway:**
  - *Clearly defining **scenario configurations** ensures **reproducibility** and **comparability** of results*

- **During Simulation**
  - During simulation run vehiclde dynamics data or logged
    - Data such as vehicle IDs, timestamps, message sizes, and object counts
  - Simulator continuously logs variables into files (e.g., CSV format)
  - Logging ensures every event is recorded for later inspection
- **After Simulation**
  - Generated files are imported into **post-processing tools** (e.g., MATLAB, Python, Excel).
  - These tools are used to:
    - **Filter** data by time, vehicle, or event type
    - **Compute metrics** such as message rate, packet size, detection accuracy, or latency.
    - **Visualize** results using different metrics

```
fp = fopen(CharsenderArrayCPM, "a");
for (std::multimap<uint32_t, senderCPM>::iterator it9 = sender_vehicles.begin(); it9 != sender_vehicles.end(); ++it9) {
    fprintf(fp, "%d, %d, %d, %f, %d, %f, %f, %f, %f, %f, %f, %f, %f, %f, %d, %f, %f ,%f ,%f, %f, %d \n", it9->first, it9->second.
    Obj_number, it9->second.CPMsize, it9->second.time, it9->second.detectedID,
            it9->second.dist_det_ID, it9->second.senderposX, it9->second.senderposY, it9->second.senderspeedX, it9->second.senderspeedY,
             it9->second.objectposX, it9->second.objectposY,
            it9->second.objectspeedX, it9->second.objectspeedY, it9->second.grouping, it9->second.position_change, it9->second.
            speed_change, it9->second.Sensor_container, it9->second.pktID, it9->second.VOI, it9->second.CPM_Size_max_bytes);
}
```

Ns-3 (c++) code for logging different data

How to Write a Paper: CSV file


```
fp = fopen(CharsenderArrayCPM, "a");
for (std::multimap<uint32_t, senderCPM>::iterator it9 = sender_vehicles.begin(); it9 != sender_vehicles.end(); ++it9) {
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    Obj_number, it9->second.CPMsize, it9->second.time, it9->second.detectedID,
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             it9->second.objectposX, it9->second.objectposY,
            it9->second.objectspeedX, it9->second.objectspeedY, it9->second.grouping, it9->second.position_change, it9->second.
            speed_change, it9->second.Sensor_container, it9->second.pktID, it9->second.VOI, it9->second.CPM_Size_max_bytes);
}
```

Ns-3 (c++) code for logging different data

```
% field(1)   : Vehicle ID
% field(2)   : Obj_number
% field(3)   : CPMsize
% field(4)   : time
```

	1	2	3	4
1	165	18	751	300.4132
2	165	19	821	300.7986
3	165	19	786	301.2143
4	165	18	751	301.5773
5	165	18	786	301.9879
6	165	19	786	302.3930
7	165	18	751	302.7936
8	165	18	786	303.2348
9	165	18	751	303.6825
10	165	18	751	304.1247
11	165	18	786	304.5139
12	165	19	786	304.9034

CSV Exported to Matlab

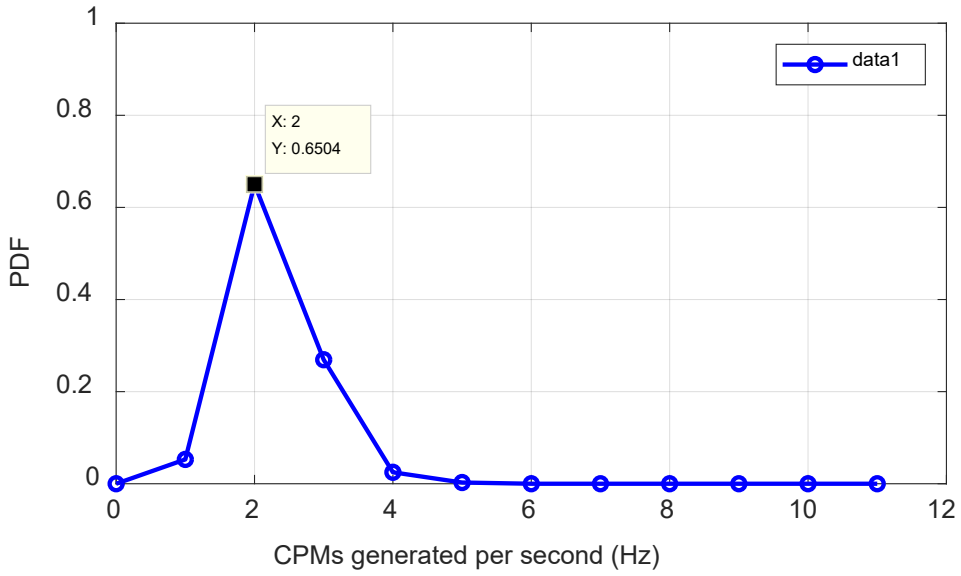
- Two Metrics are computed using different fields
  - How many messages are generated by vehicle for each second
  - How many objects (vehicles) are detected by the in-vehicle sensor

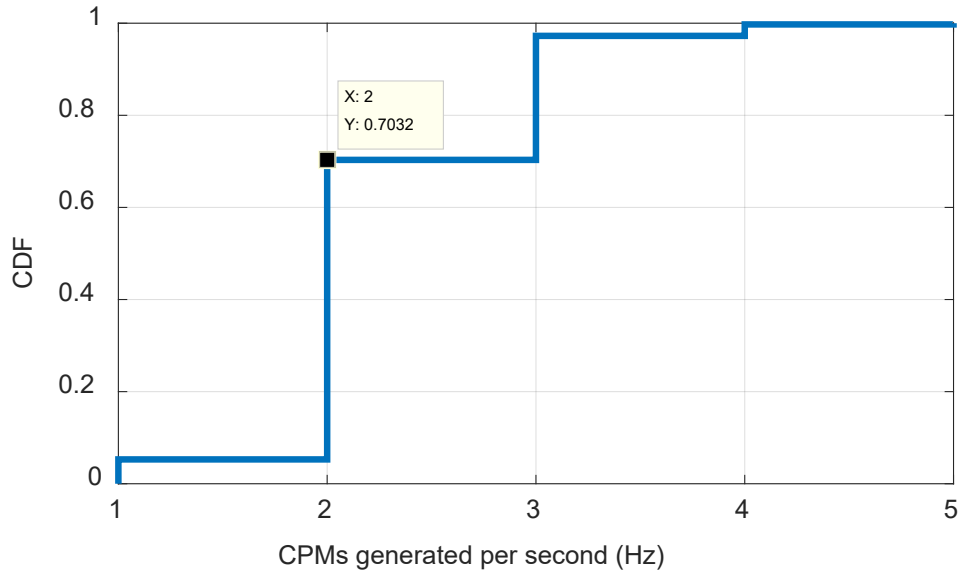
Metrics

- Many engineering quantities are random:
  - received signal power, thermal noise, packet arrival times, sensor range error
- We can't predict an exact value, but we *can* describe the pattern
- For continuous X: P(X = x) = 0 for every single value x
  - There are infinitely many values, so no single point can carry positive probability
  - probability that a received power is exactly -20 dBm, to infinitely many decimal places? Zero.
- Meaningful question: probability of falling in an **interval** $P(a \leq X \leq b)$

How to Write a Paper: Message (CPMs) Generated per second

- PDF (Probability Density Function)
  - The peak in the curve indicates the most common or likely CPM rate (e.g., 2 Hz)
  - Useful for identifying dominant behaviors or most frequent values among vehicles
- CDF (Cumulative Distribution Function)
  - The curve starts at 0 and ends at 1, increasing as the CPM rate grows
  - Enables direct probability interpretation e.g., 70 % of vehicles have CPM ≤ 2 Hz





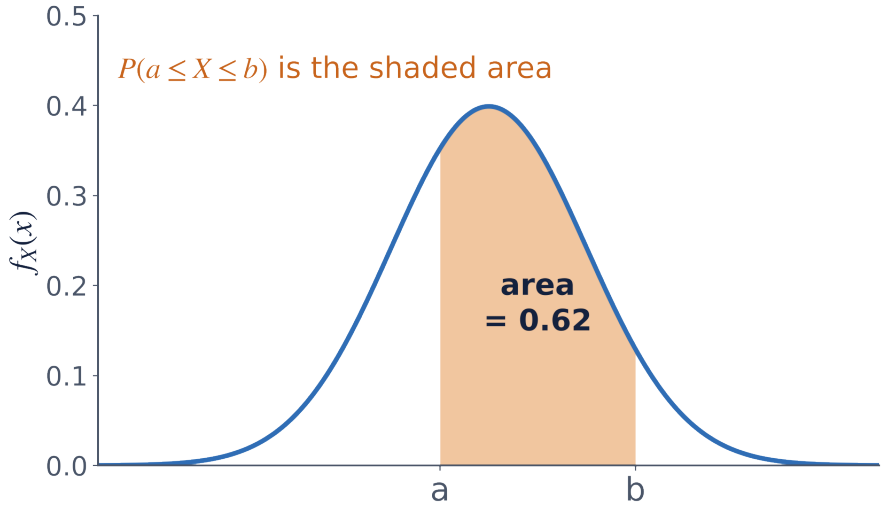
Probability Density Function (PDF)

- PDF is the function whose area gives probability
  - To find the chance that X lies between a and b, integrate the PDF over that interval

$$P(a \le X \le b) = \int_a^b f_X(x)\,dx$$

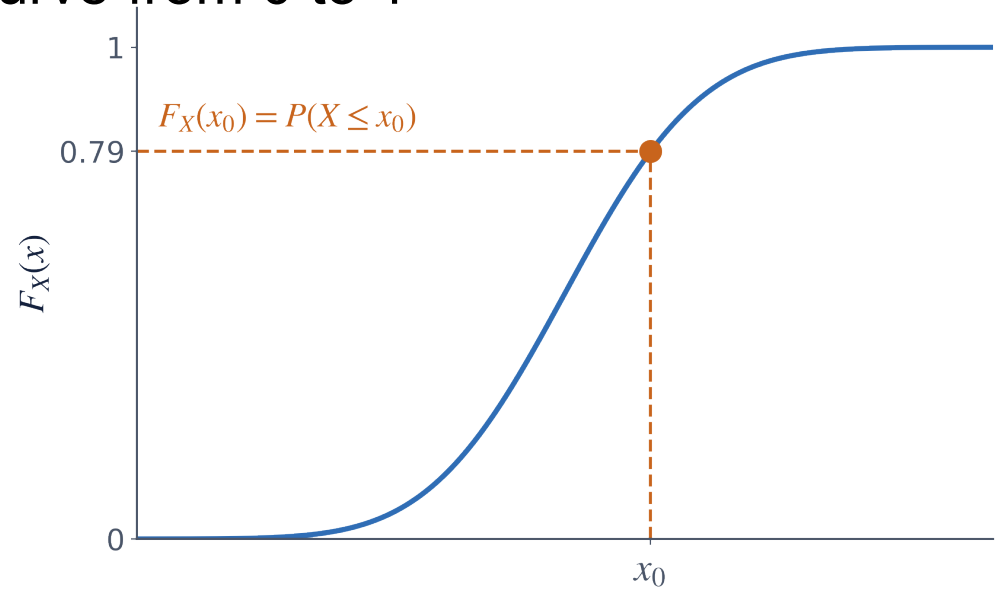
- It must never be negative, and its total area must be one

$$f_X(x) \ge 0, \qquad \int_{-\infty}^{\infty} f_X(x)\,dx = 1$$



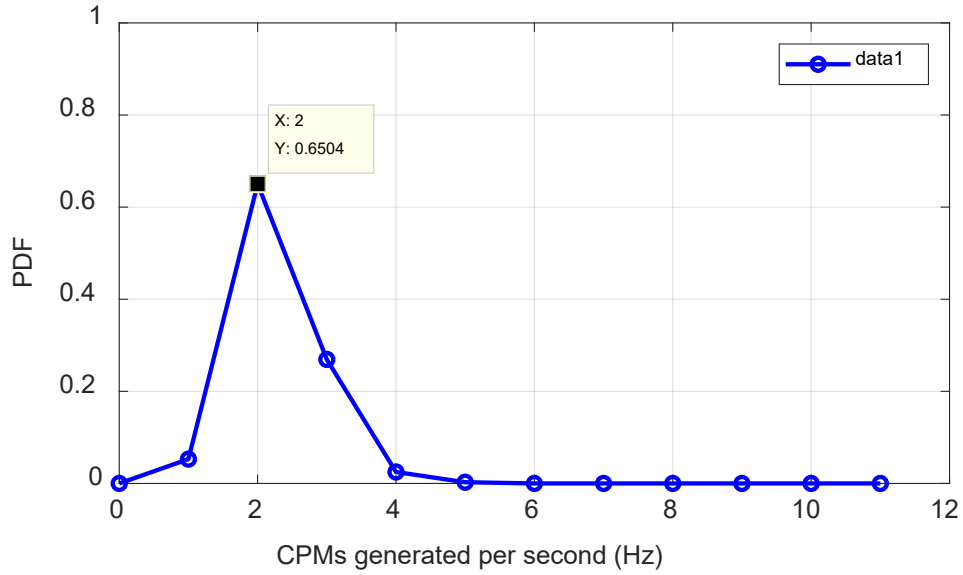
Cumulative Distribution Function (CDF)

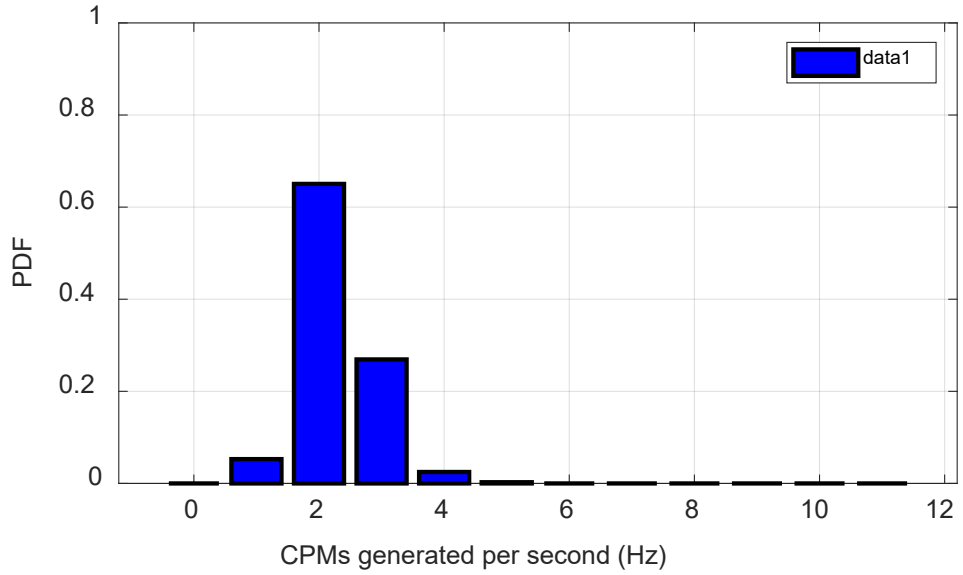
- CDF answers what's the probability that X is less than or equal to x?
- $$F_X(x) = P(X \leq x)$$
- Probability that X is at most x, accumulated from $-\infty$
  - Start from the far-left end of all possible values and accumulate all probability up to (x)
- Defined for every random variable: discrete, continuous or mixed
- For continuous variables, usually a smooth S-curve from 0 to 1



79% of the observations are less than or equal to (x0)

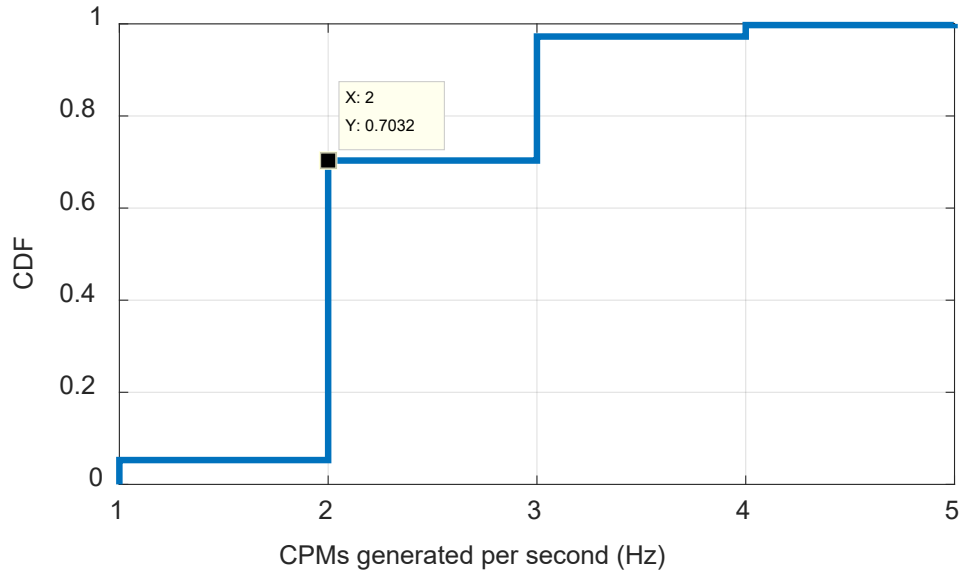
- **PDF (Line or Histogram):**
  - The smooth line (PDF curve) provides a **continuous view** of the same distribution **(left fig)**
    - Useful for identifying the **dominant CPM generation rate** (e.g., peak at 2 Hz)
  - The histogram highlights how data values are distributed across bins **(right fig)**
    - Shows **frequency concentration**

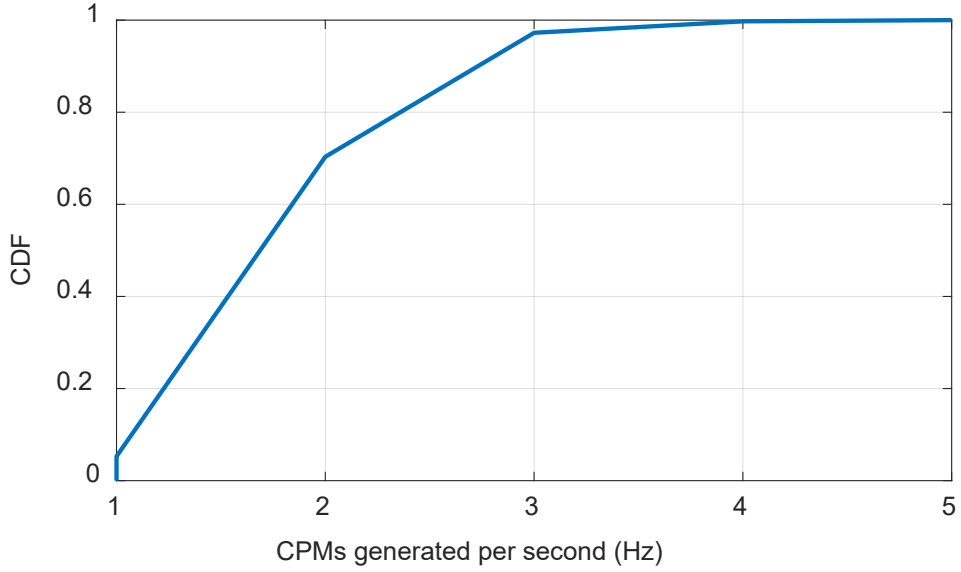




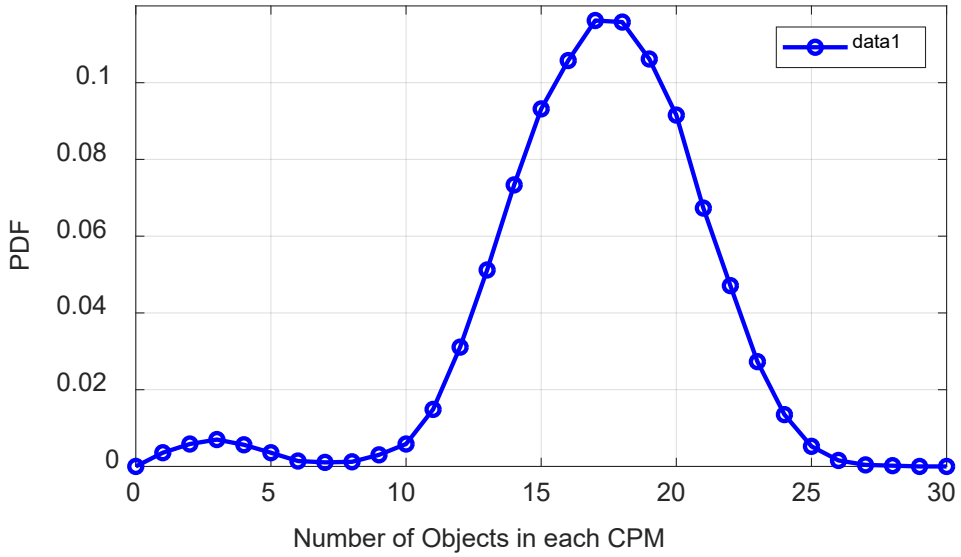
How to Write a Paper: Message (CPMs) Generated per second

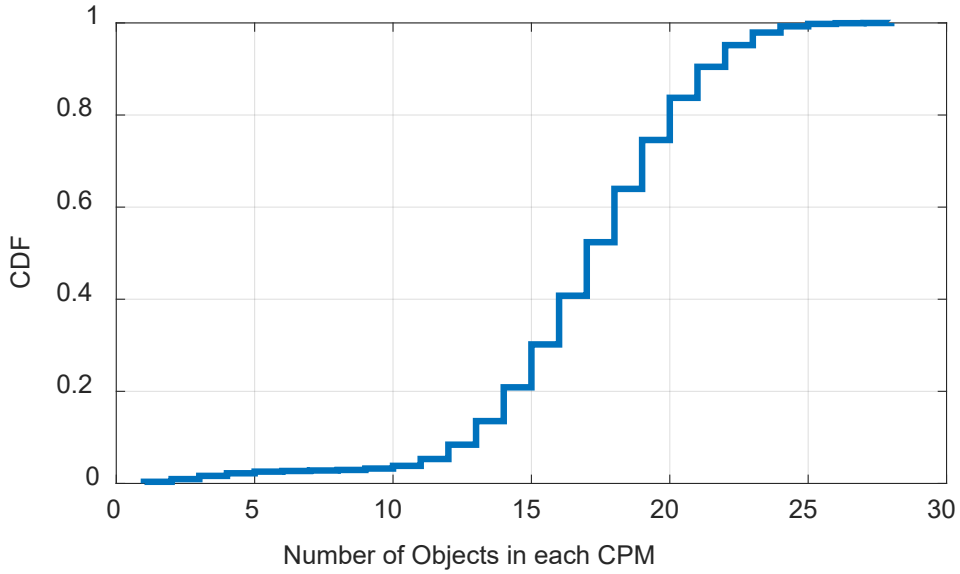
- **CDF (Step vs. Smooth):**
  - **Step CDF** clearly shows discrete jumps in cumulative probability **(left fig)**
    - Suitable for **event-based datasets** like CPM generation
  - **Smooth CDF** approximates a continuous trend **(right fig)**
    - Visualize **overall cumulative behavior** and estimate **percentiles** (e.g., 70 % of vehicles ≤ 2 Hz).





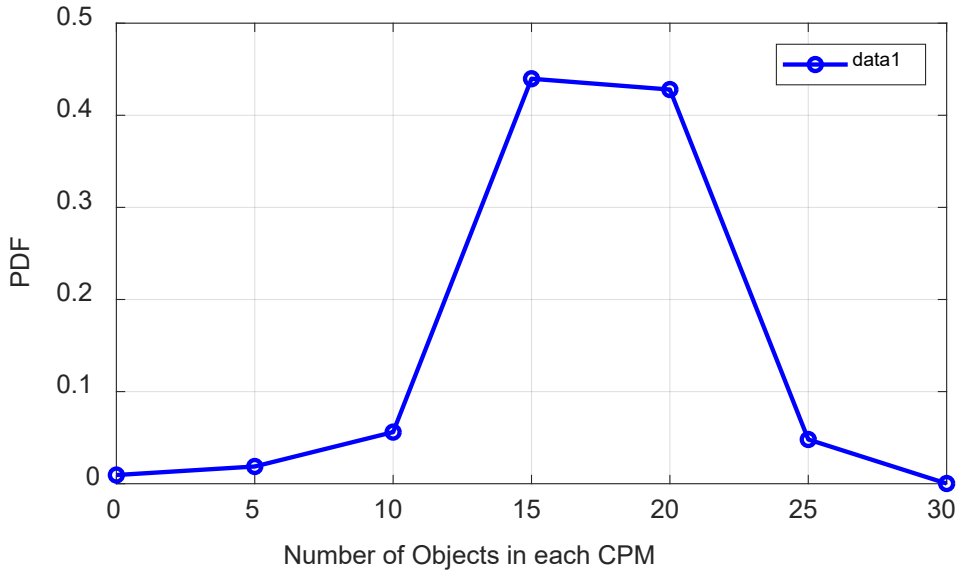
- **PDF (Probability Density Function)**
  - Represents how frequently different **numbers of detected objects** occur in each CPM
  - The **peak of the curve** shows the most common detection count (e.g., ~18 objects per CPM)
  - Useful to identify **dominant perception trends**
- **CDF (Cumulative Distribution Function)**
  - Illustrates the accumulated probability of detecting up to a certain number of objects.
  - Example: From the curve, about 80% of CPMs contain ≤ 20 detected objects.
  - Helps interpret overall sensor performance and variability in perception capability among vehicles

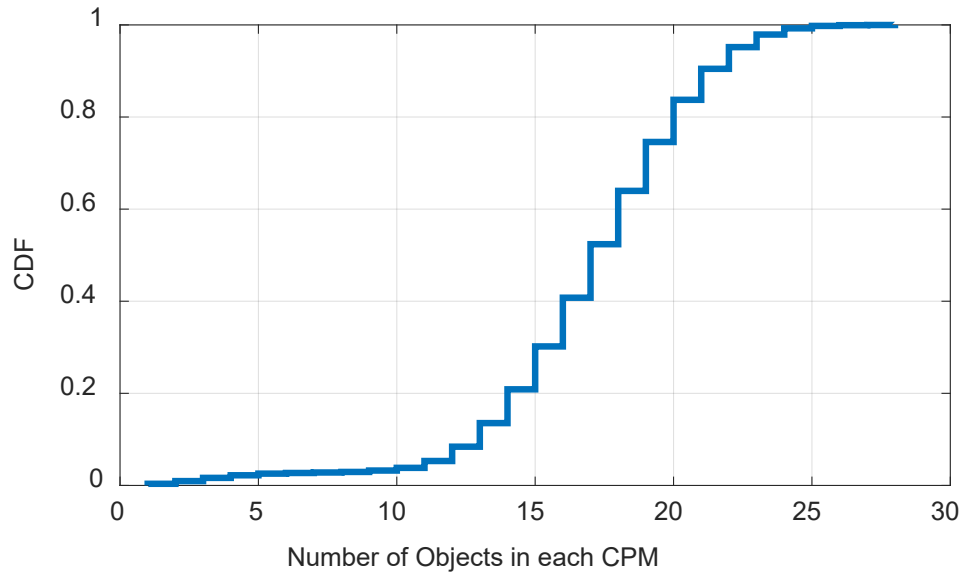




How to Write a Paper: Objects detected by the vehicle sensor

- **PDF (Probability Density Function)**
  - The **bin size** determines how data points are grouped when computing the PDF.
  - **Smaller bins** capture fine variations but can make the curve noisy or irregular
  - **Larger bins** smooth the curve but may hide smaller patterns or local peaks
  - **PDF shape** interpretation should focus on **overall trends** rather than small fluctuations
- **CDF (Cumulative Distribution Function)**
  - The CDF is **independent of bin size**, as it represents the **cumulative probability** up to a given value
  - Changing bin size in the PDF only alters how probabilities are distributed locally
  - **Total accumulation** in the CDF remains consistent

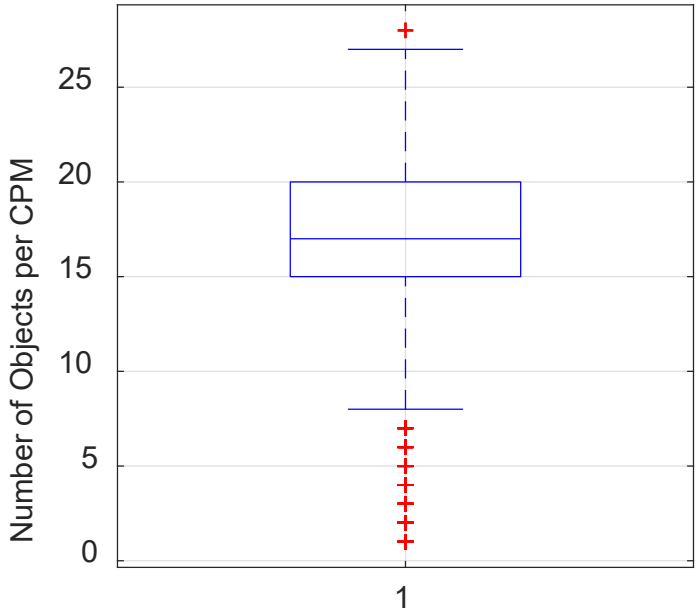


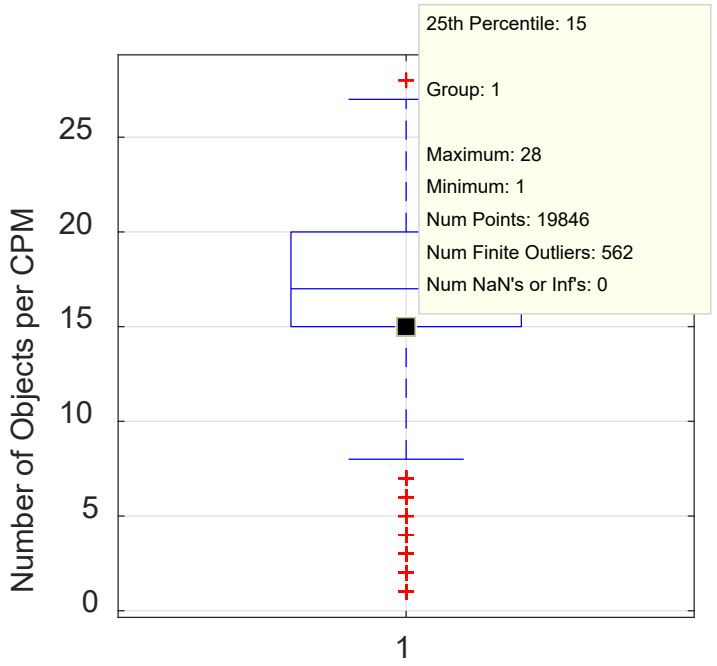


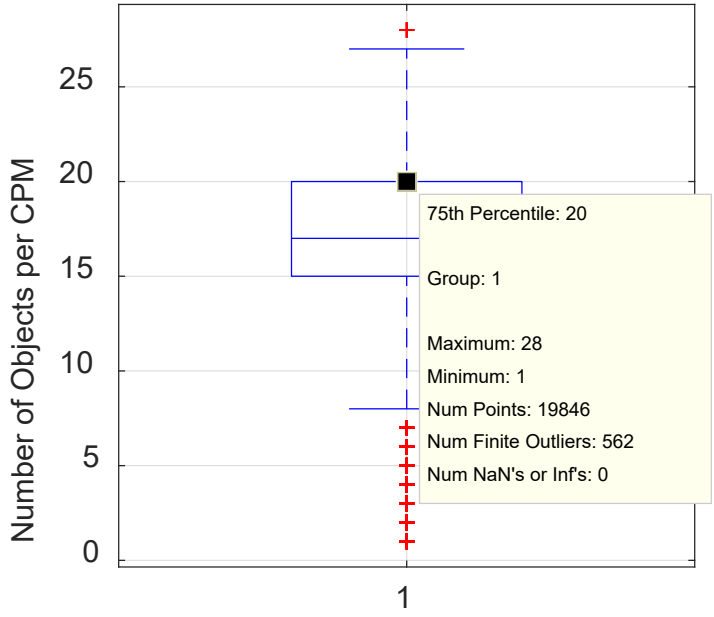
How to Write a Paper: Objects detected by the vehicle sensor

- Box Plot: spread, median, quartiles, and outliers

Element	Represents	Meaning in the Figure
Bottom whisker	Minimum (excluding outliers)	Smallest observed number of objects per CPM
Bottom of box	25th percentile (Q1= 15)	25% of CPMs have fewer than this many objects
Line inside box	50th percentile (median)	Half of the CPMs are below this value
Top of box	75th percentile (Q3 = 20)	75% of CPMs have ≤ 20 objects
Top whisker	Maximum (excluding outliers)	Largest CPM object count within normal range
Red +	Outliers	CPMs that have unusually few or many detected objects







Assignment 2: Compute different metrics

- Use the CSV OR matlab file to compute different metrics
  - The log field information is provided

Field #	Variable	Description
1	Vehicle ID	Transmitting vehicle identifier
2	Obj_number	Number of detected objects by the vehicle sensor
3	CPMsize	Size of the CPM message (bytes)
4	time	Simulation timestamp (s)
5	detectedID	ID of detected object/vehicle
6	dist_det_ID	Distance between sender and detected object (m)
7	senderposX	Sender position X-coordinate (m)
8	senderposY	Sender position Y-coordinate (m)
9	senderspeedX	Sender velocity X-component (m/s)
10	senderspeedY	Sender velocity Y-component (m/s)
11	objectposX	Detected object position X (m)
12	objectposY	Detected object position Y (m)
13	objectspeedX	Detected object velocity X (m/s)
14	objectspeedY	Detected object velocity Y (m/s)
15	grouping	Grouping indicator (1 = grouped, 0 = single)
16	position_change	Position change flag since last CPM
17	speed_change	Speed change flag since last CPM
18	Sensor_container	Type of sensor data (camera/radar/lidar)
19	pktID	Unique CPM packet identifier
20	VOI	Value of Information metric
21	CPM_Size_max_bytes	Maximum CPM size (bytes)

Do not consider the red fields

Thank you for your attention

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